Gain Scheduled PI Speed and Current Controllers for the Vector Control of PMSM

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***Abstract*—Permanent magnet synchronous motors (PMSM) are used in various applications in industrial, domestic, and aerospace sectors. Vector control of PMSM provides better dynamic and steady state performance. This method of control consists of one speed and two (d and q-axes) current controllers, which can be proportional plus integral (PI), fuzzy or neural network based. Gain scheduled PI controllers have been used as the speed controllers of PMSM with benefits such as automated tuning, ease of computation and lesser execution time. In the work reported in this paper, apart from the speed controller, gain scheduled PI controllers are also employed for the two current controllers for the vector control of the PMSM. Three different functions namely, the exponential, hyperbolic tangent and polynomial S function have been utilized to determine the integral gain during each sampling instant in all the three controllers. The control algorithms have been simulated and the hardware implementation has been carried out with TMS320F2812 digital signal processor. From the simulations as well as the hardware results, it is evident that the performance of the vector controlled PMSM drive is superior with the gain scheduling based on hyperbolic tangent and polynomial S function for the determination of integral gain.**

***Index Terms*—Controller, motor, permanent magnet motor, brushless dc motor, PMSM, speed control, vector control.**

I. Introduction

THE developments in the field of permanent magnet materials and power converters and availability of high speed digital signal processors (DSP) combined with the advantages of higher efficiency, low rotor inertia, high power density and robust construction [1]-[4] has resulted in wide interest in the PMSM for applications in robotic actuators, computer disk drives, domestic appliances, heat ventilating air conditioning, robotics, electric vehicles, machine tools, servos, and aerospace mechanisms [5]-[6]. The developments in the field of fast DSP and field programmable gate arrays (FPGA) have facilitated the implementation of complex control algorithms in these motors [2], [4] and [7].

Vector control of the PMSM results in better dynamic performance [2]-[3]. Traditionally, PI controllers are preferred as speed and current controllers [8]-[9]. Due to fixed proportional gain and integral time constants, the PI controllers are affected by parameter variations, load disturbances and speed variations; these can be overcome by the fuzzy logic controller [10], which do not require a mathematical model and are based on the linguistic rules obtained from the experience of the system operator and provide robust performance [2]. Compared to the PI controller, fuzzy logic control algorithm is computationally extensive and time consuming. The performance of the fuzzy logic controller is superior only under transient conditions, while the PI controller performance is better under steady state condition. A simple switching algorithm can be implemented to combine the advantages of these two types of controllers [11]; but the performance of such a hybrid controller has to depend on the switching function formed for assigning weights for the outputs of the PI and fuzzy controllers.

Self tuned PI controllers with fuzzy, genetic algorithm and neural network based tuning algorithm [12]-[14] suffer from intensive computational time and increased complexity. Instead, a simpler gain scheduled PI speed controller for permanent magnet brushless DC (PM BLDC) motor has been reported in which the automatic tuning is possible, and the factor α, which is used to determine the integral gain, is determined with an exponential function [15]. The factor α can also be determined with hyperbolic tangent function [16]. Guney et al [17] present the vector control of PMSM with fuzzy speed and current controllers, but this necessitates large execution time.

In order to reduce the computational burden and also to improve the performance of speed and current controllers, a new control strategy for the vector control of PMSM with gain scheduled PI speed and d-q axes current controllers which results in a drive, free of controller tuning requirements has been presented in this paper. Gain scheduled PI controller has been implemented with integral gain being determined with an exponential function in case-1, with a hyperbolic tangent function in case-2, and with a polynomial S function (often used as a membership function in fuzzy logic [2]) in case-3.

II. Vector Control of PMSM with Gain Scheduled PI Speed and D-Q Axes Current Controllers

A. Vector Control of PMSM

With vector control, PMSM is made to emulate the superior dynamic characteristics of dc motor with appropriate field orientation [2] and [18]. Not only the magnitude and

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frequency, but also the phase of the currents is controlled. The stator currents are resolved into the torque producing and flux producing components with the co-ordinate transformation. These two current components are controlled independently as per the torque demand. Vector controlled PMSM when operated in constant torque region, can be operated with constant torque angle control. Under constant torque angle control, the torque angle is maintained at 90 degrees. Only the quadrature component of the current is present, with the direct axis component being zero [2]-[3]. The mathematical model for the PMSM in the synchronously rotating reference frame aligned with the rotor flux linkages is expressed in (1)-(5), [3].

$$v_d = i_d R + p\lambda_d - \omega_e \lambda_q \quad (1)$$

$$v_q = i_q R + p\lambda_q + \omega_e \lambda_d \quad (2)$$

$$\lambda_d = L_d i_d + \lambda_{af} \quad (3)$$

$$\lambda_q = L_q i_q \quad (4)$$

$$T_e = 3P\ [\lambda_{af} i_q + (L_d - L_q) i_d i_q]/2 \quad (5)$$

Where, v_d *and* v_q are the instantaneous d-q axes voltages, i_d and i_q are the d-q axes stator currents, L_d *and* L_q are d-q axes inductances, and λ_d *and* λ_q are the d-q axes stator flux linkages respectively, λ_{af} is the stator flux linkage due to the rotor magnets, R is the stator resistance, P is the number of pole pairs and T_e is the developed electromagnetic torque. With i_d being held at zero, torque is controlled by the q-axis current.

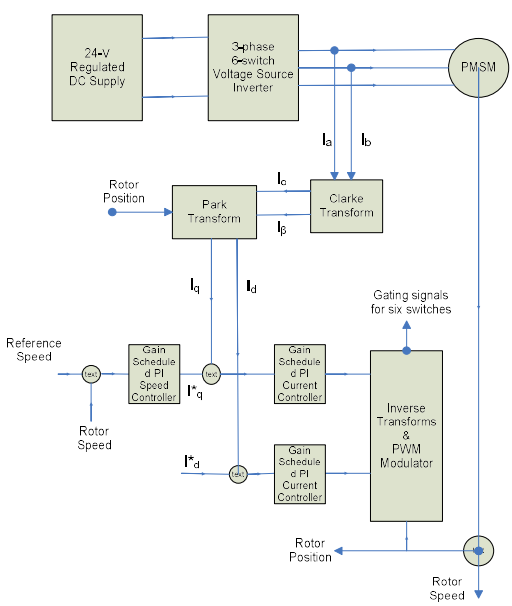

Fig. 1. Schematic diagram of the vector control of PMSM with gain scheduled PI speed and current controllers

Fig. 1 shows the block diagram for vector control of the PMSM. The A and B phase currents are sensed by means of current sensors and then a two-phase to three-phase transformation is carried out with Clarke and Park transforms. In order to control the mechanical speed of the motor, an outer speed loop is implemented. The actual motor speed and position is sensed with the help of incremental encoder with index pulse and quadrature encoder pulse circuit (QEP). The actual and the estimated speeds are compared and speed error is processed through a speed controller, which provides the reference q-axis current. Under constant angle control, for operation in constant torque region the reference value of i_d^* is held at zero. This leads to an increase in motor efficiency and maximizes the torque per ampere [2]-[3]. The d-axis and q-axis current errors are processed through the individual current controllers to provide the reference value of the d and q-axes voltages. As per the modulation strategy selected, appropriate inverse transforms are carried out and the voltages are applied across the motor terminals with appropriate switching of the inverter switches.

B. Gain Scheduled PI controller

The block diagram of gain scheduled PI controller is shown in fig.2. The error input is processed by the gain scheduling algorithm to determine the values of proportional and integral gains. At the instant of starting the proportional gain is derived such that the processed error generates a control signal which stabilizes the error, and under the steady state condition the integral gain is utilized to eliminate the steady state error [16]. When the error input is high, the proportional gain is the maximum, and it reduces as the error decreases. The integral gain comes into prominence to eliminate the steady state error.

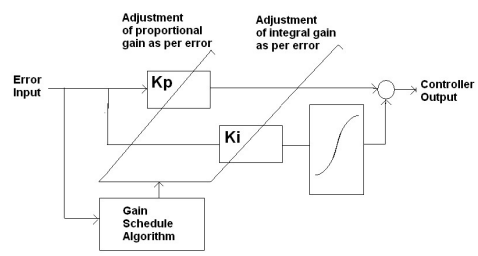

Fig. 2. Block diagram of gain scheduled PI speed and current controller

The proportional gain can be expressed as a function of the error input in discrete form, as shown in (6) [15]-[16].

$$K_p[n] = K_{p\max} - (K_{p\max} - K_{p\min}) \exp^{-(G|err[n]|)} \quad (6)$$

Where $K_p[n]$ is the proportional gain at the nth sampling instant, $K_{p\max}$ and $K_{p\min}$ are the maximum and minimum values of the proportional gain, $err[n]$ is the error input to the gain scheduled controller at the nth sampling instant, and G is the constant gain which determines the rate of variation of the proportional gain. Under transient conditions, the error is large and the large proportional gain is utilized to drive the process towards the steady state condition.

Exponential Function based Integral Gain Scheduling

The integral gain can be determined at each sampling instant with exponential function as shown in (7) [15].

$$K_i[n] = K_{i\max} \exp^{-(G|err[n]|)} \quad (7)$$

Where $K_i[n]$ is the integral gain at the nth sampling instant, $K_{i\max}$ is the maximum value of the integral gain. Constant gain G determines the rate of change of the integral gain as per the error input. With the error being large, the integral gain is reduced to a minimum. Under steady state condition, the error is small leading to the utilization of a larger integral gain for elimination of the error by the integral action.

Hyperbolic Tangent Function based Integral Gain Scheduling

The hyperbolic tangent function can also be utilized for the derivation of the integral gain as shown in (8) [16].

$$K_i[n] = (1-\alpha[n])K_{p\max} \quad (8)$$

Where, $\alpha[n]$ is the gain which determines the rate of change of the integral gain and is expressed as (9).

$$\alpha[n] = \tanh(\beta[n]) \quad (9)$$

Where,

$$\beta[n] = |e[n]| - \varepsilon \quad \text{if} \quad |e[n]| \geq \varepsilon > 0 \quad (10)$$

and

$$\beta[n] = 0 \quad \text{if} \quad |e[n]| < \varepsilon \quad (11)$$

Polynomial S Function based Integral Gain Scheduling

With the hyperbolic tangent function replaced by the polynomial S function, the rate of variation of the integral gains near the zero and the maximum input error is slower, which is better.

III. Simulation Results

Simulations have been carried out for the three cases of gain scheduling techniques for the vector control on a 24 V, 100W, 7A, 4000 rpm PMSM with a load torque of 0.02 Nm.

Simulation results for case-1, i.e., with the exponential function for the determination of the integral gains are shown in fig. 3-4. As observed from fig. 3, the motor speed oscillates at a very low frequency before settling down. The peak overshoot is 53 rpm over the reference value of 4000 rpm. The line current is fairly sinusoidal as shown in fig. 4; the maximum current being 8.041 A at starting. Further investigations on the oscillations in the motor speed revealed the cause for this as the oscillations in speed and q-axis current controller outputs, which are inherent with the exponential function based integral gain determination scheme. Thus, the performance of the vector controlled PMSM is not that good in this case.

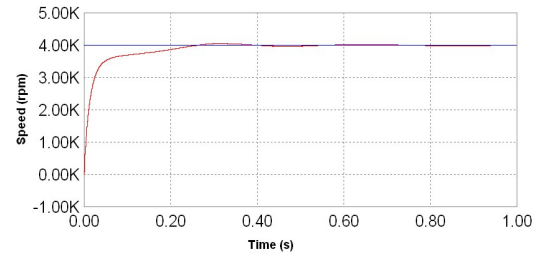

Fig. 3. Speed of the PMSM with gain scheduled PI controllers for case-1

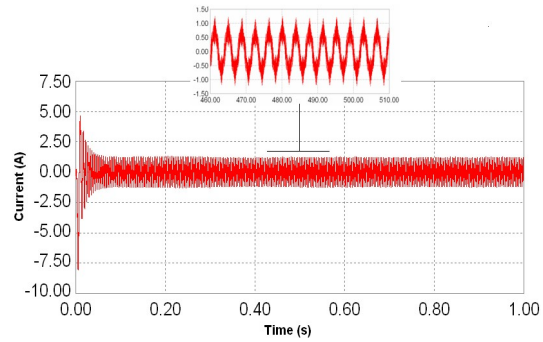

Fig. 4. Line current for case-1

Simulation results for the case-2, i.e. with the integral gain determination based on hyperbolic tangent function, are given in fig. 6-7. From fig. 5, it can be observed that there are no oscillations, overshoots or dips in the motor speed. The settling time is 0.360 s. The maximum current at starting is 3.921 A only against the rated line current of 7 A as shown in fig. 6. In this case also, the line current is fairly sinusoidal.

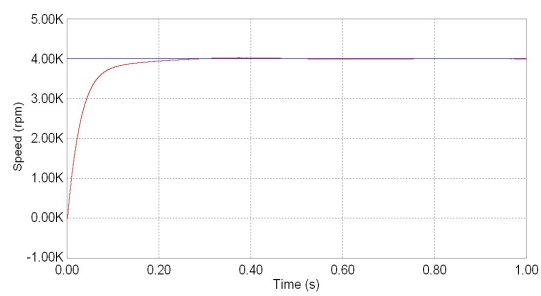

Fig. 5. Speed of the PMSM with gain scheduled PI controllers for case-2

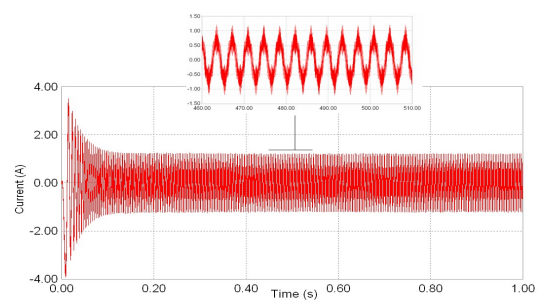

Fig. 6. Line current for case-2

The variations of the calculated values of the proportional and integral gains with respect to time are shown in fig. 7. The proportional gain is at its maximum value, 1 unit, at starting

and then reduces as the speed increases, with existence of a small oscillation, and settles down at 0.102 units. On the other hand, the integral gain is at its minimum at starting, 1.197 units, and gradually increases with the increase in speed and attains a maximum value of 5 units as the motor speed settles at its reference value of 4000 rpm. Thus, the performance of the motor with this scheme is better when compared with that of the case-1.

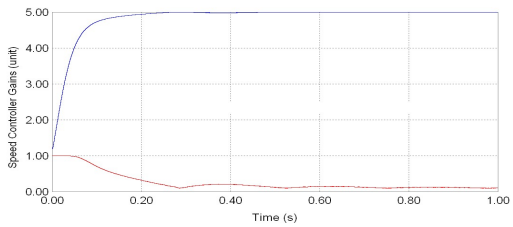

Fig. 7. Gain scheduled PI speed controller gains for the vector control of PMSM

In case-3, where polynomial S function is used to determine the integral gain, the results of the simulation shown in fig. 8-10 reveal that in this case too, there are no over or undershoots in speed and he maximum line current at the starting is 3.865 A only. In this case also, the line current is fairly sinusoidal. Fig. 10 shows the variation of proportional and integral gains with time, which are similar as to that in case- 2. Proportional gain has peak value of 1unit, at the instant of starting and reduces with the increase in speed and settles at 0.125 units. The integral gain is 0.6005 units at the instant of starting and increases to 4.404, lesser than in case-2, as the rotor approaches the reference speed.

On comparison among the three cases, it can be observed that the cases-2 and 3 are definitely better than case-1, and among the cases-2 and 3, case-3 is marginally better owing to the slight reduction in the maximum current.

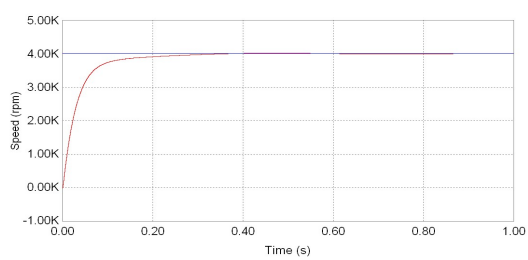

Fig. 8. Speed of the PMSM with gain scheduled PI controllers for case-3

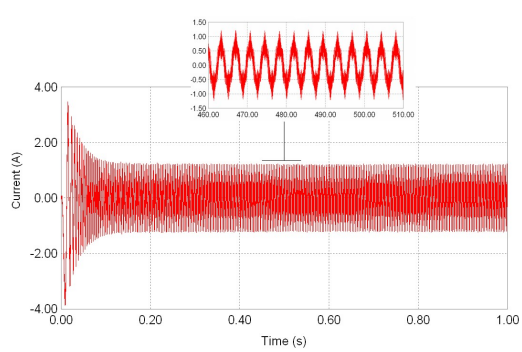

Fig. 9. Line current for case-3

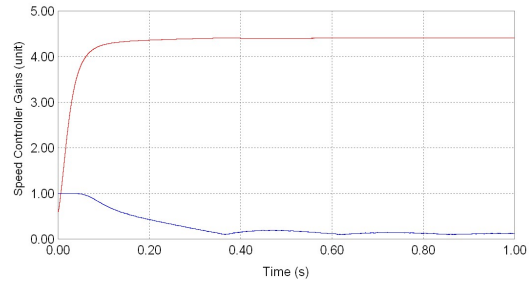

Fig. 10. Gain scheduled PI speed controller gains for the vector control of PMSM for case-3

IV. Hardware Implementation and Results

The three cases have been implemented on a 100W, 7A, PMSM with TMS320F2812 DSP at no load. The hardware setup is provided with a protective relay circuit, which trips when the current exceeds 7 A. The reference speed in each case is chosen to be 1500 rpm, with a regulated DC bus voltage of 15 V.

The measured speed and line current in case-1 are shown in fig. 11-13. Fig. 11 shows the motor speed in which a steady state error is observed. The steady state rms current is 1.646.

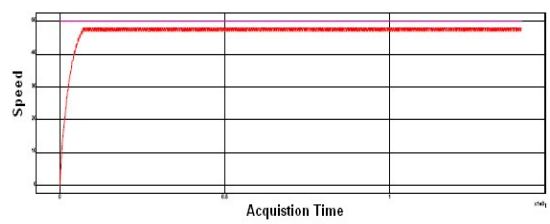

Fig. 11. Measured speed of the PMSM with gain scheduled PI controllers for reference speed of 1500 rpm for case-1

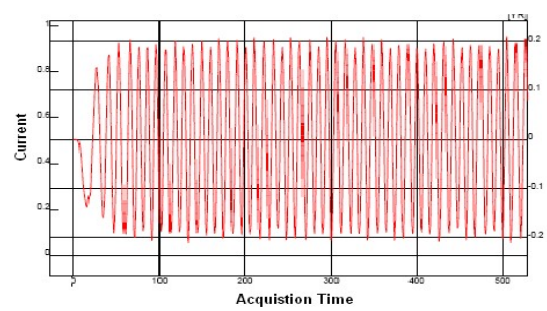

Fig. 12. Measured line current for case-1

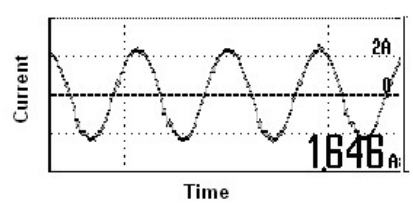

Fig. 13. Measured steady state line current for case-1

The measured speed and line current in case-2 are shown in fig. 14-18. Motor attains the reference speed without any oscillations or overshoots and there is no steady state error in this case. The steady state rms current is 1.082.

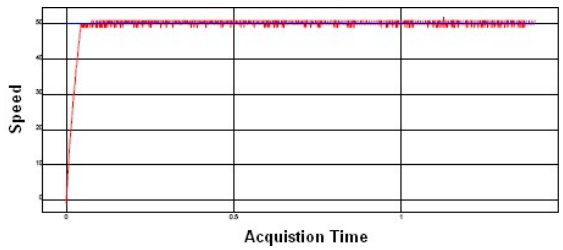

Fig. 14. Measured speed of the PMSM with gain scheduled PI controllers for reference speed of 1500 rpm for case-2

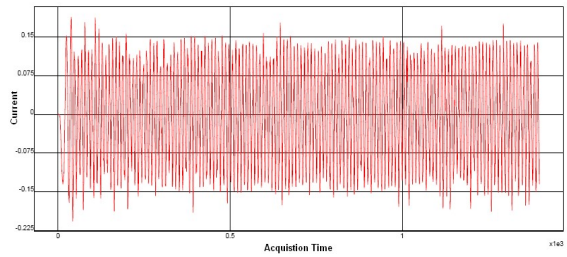

Fig. 15. Measured line current for case-2

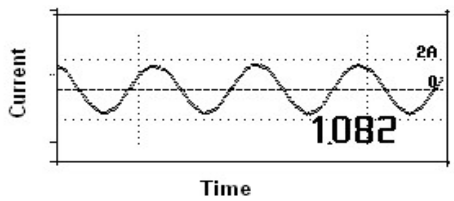

Fig. 16. Measured steady state line current for case-2

Fig. 17 shows the output of gain scheduled PI speed controller. The speed controller is initially just above 10 units as the reference speed is being attained, and then sharply falls to 4 units and then it slowly reduces with the variations being contained within a band. Fig. 18 shows the gain scheduled PI q-axis current controller output. The output of q-axis current controller increases to an average value of 22.5 units as the motor attains the reference speed. There is no sudden overshoot, oscillations or dips in the output of the gain scheduled PI controllers which results better performance of the speed and current responses of the motor in terms of steady state error in speed, steady state current magnitude and the harmonic content in the current.

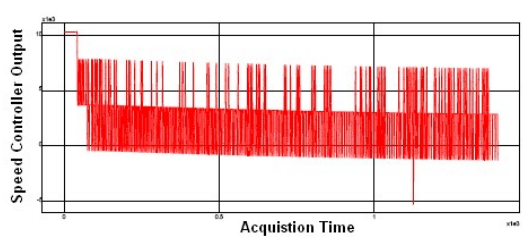

Fig. 17. Gain scheduled PI speed controller output for vector controlled PMSM for case-2

The measured speed and line current in case-3 are shown in fig. 19-23. Motor attains the reference speed without any oscillations or overshoots in this case also. The steady state error in the speed is much lesser than the case-1. The steady state rms current is 0.974 A.

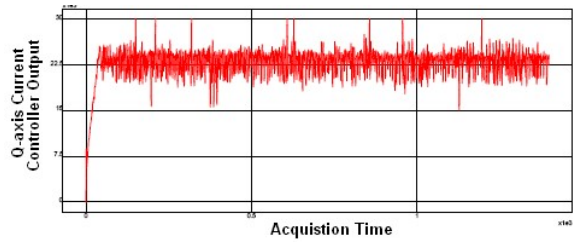

Fig. 18. Gain scheduled PI q-axis current controller output for vector controlled PMSM for case-2

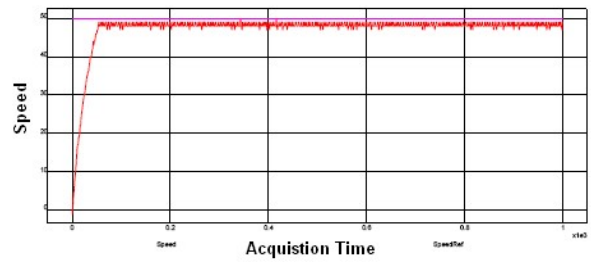

Fig. 19. Measured speed of the PMSM with gain scheduled PI controllers for reference speed of 1500 rpm for case-3

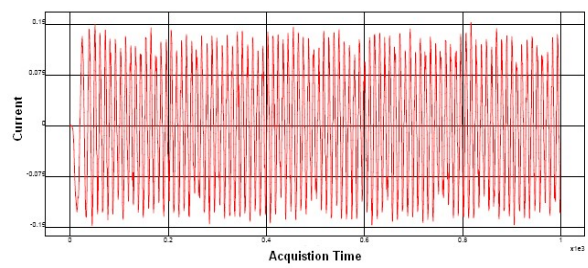

Fig. 20. Measured line current for case-3

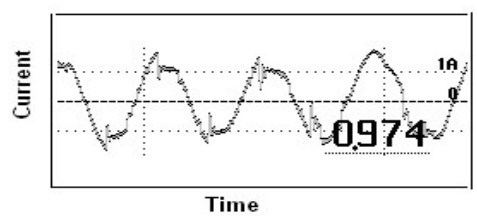

Fig. 21. Measured steady state line current for case-3

Fig. 22 shows the output of the gain scheduled PI speed controller. The speed controller is initially just above 10 units as the reference speed is being attained, and then sharply falls to an average value of 7 units. Fig. 23 shows the gain scheduled PI q-axis current controller output, where the increases to an average value of 22 units as the motor attains the reference speed. There is no sudden overshoot, oscillations or dips in the output of the gain scheduled PI controllers which results in better performance drive in terms of the magnitude of the steady state current; the least value of the steady state current is in this case. The harmonic content in the measured steady state current can further be reduced by proper tuning of the limits of the gains of the gain scheduling algorithms.

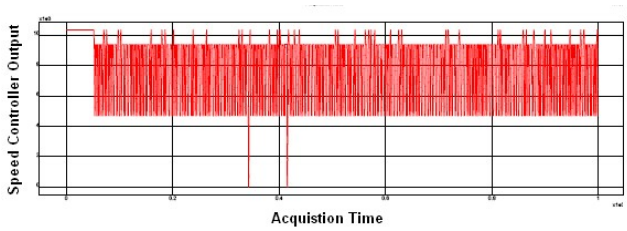

Fig. 22. Gain scheduled PI speed controller output for vector controlled PMSM for case-3

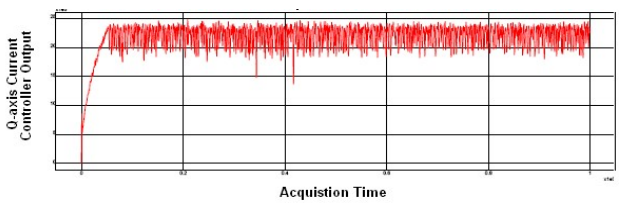

Fig. 23. Gain scheduled PI q-axis current controller output for vector controlled PMSM for case-3

TABLE I
COMPARISON OF DRIVE PERFORMANCE FOR THE THREE CASES OF TECHNIQUES FOR THE DETERMINATION OF THE INTEGRAL GAIN

Case	Simulation results (for 24V, 0.02 Nm load, 400 rpm speed)		Hardware results (for 15 V, no load, 1500 rpm)
	Settling Time (s)	Maximum current at starting (A)	Steady state rms current (A)
1	>1	8.04062	1.646
2	0.35957	3.9212	1.082
3	0.393957	3.86492	0.974

Table I presents the comparison of the drive performance for three cases of techniques for the determination of the integral gains. It can be observed from this table that the cases-2 and 3 are definitely better than case-1. The case-3 is better than case-2 in terms of the magnitude of the starting current and also the steady state currents.

V. CONCLUSION

Simulation and hardware implementation of vector control of PMSM has been presented with three different functions for the determination of integral gain. With the presented control strategies, which can be achieved with simple algorithms, that are not computationally intensive or needing any corrective reference or adjustment algorithms, the drive is free from the requirement of tuning PI controllers. From the simulation and hardware results, it is concluded that the cases-2 and 3 are better than casaes-1 in terms of steady state error and oscillations in speed, and the magnitude of the starting and steady state currents. The case-3, which is with the polynomial S function, a novel scheme reported in this work, seems to be a better choice in terms of the state currents and the maximum starting current. With proper tuning of the limits of the gains in the gain scheduling algorithms, the performance of the vector controlled PMSM using this novel scheme can further be improved in terms of harmonic contents in the line current and steady state error in speed.

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